

ePTW SYSTEM REBUILD

ePTW Rebuild — Requirements Review

Proposed features and logic, presented for end-user confirmation

Requirements Confirmation Deck

How to Read This Deck

Each section states the proposal, then asks what needs confirming

PROPOSED BEHAVIOUR

A plain statement of how the rebuilt system will work — not a pitch, just the facts as currently designed.

CONFIRM WITH END USER

Specific open questions where we need your input before this is finalised.

Agenda

24 sections

1. Purpose of the Rebuild

2. Proposed Scope

3. User Roles & Permission Matrix

4. Login & Account Management

5. Permit Type & Template Mgmt

6. Permit Creation Process

7. Permit Numbering Logic

8. Parent & Child Permit Logic

9. Review & Approval Workflow

10. Permit Status & Lifecycle

11. Field/Button Enable-Disable

12. Approved Permit Amendment

13. Permit Extension & Expiry

14. Closure, Cancellation, Revocation

15. Email & System Notifications

16. Permit Search & Listings

17. Dashboards & Reports

18. Permit PDF Generation

19. Audit Trail & Change History

20. Administration & Configuration

21. Security & System Controls

22. Proposed User Journey

23. Key Changes from Existing ePTW

24. Items Requiring Confirmation

01

Purpose of the ePTW Rebuild

Why this system is being rebuilt

Why the System Is Being Rebuilt

Four categories of problem found in the current system

PROPOSED BEHAVIOUR

- Safety gaps — the approval sequence is enforced by the screen, not the server; a rejected permit can be resubmitted into a status where nobody can approve it
- Data integrity gaps — a company typo silently strands a permit; a renamed template field silently orphans historical data with no error
- Security gaps — an unguarded discipline API; a shared plaintext admin password reset with no role check
- Missing capability — no live safety monitoring (gas readings), no real audit trail, no way for admins to edit a site or discipline

CONFIRM WITH END USER

? *Does this framing match leadership's understanding of why the rebuild was approved?*

Main Improvement Objectives

What the rebuild is designed to achieve

PROPOSED BEHAVIOUR

- A fixed, server-enforced approval chain that cannot be bypassed or stalled
- Structured master data (Company, Site, Discipline) replacing free text — eliminates silent mismatches
- A genuine audit trail — every permit action and every configuration change, attributable and permanent
- New safety-monitoring capability for high-risk work (Assessor readings) that doesn't exist today
- A simplified, consistent role model — 6 roles instead of 8, each with a distinct purpose
- Consistent, purposeful reporting — 6 dashboards that each answer one clear question

Purpose of This Review Session

What we need from you in this session

PROPOSED BEHAVIOUR

- Every feature in this deck has already been designed in detail through prior working sessions
- This session is to CONFIRM the logic matches real operational needs — not to design from scratch
- Where a decision was made, it's stated as a fact. Where something is still open, it's flagged for your input
- Section 24 collects every open question in one place as a final checklist



02

Proposed Scope

What's in, what's kept, what's new, what's out

Scope Overview

Category	Examples
Retained from today	Company/permit visibility model, drawn signatures for contractor declaration & closure, email-only notifications, Carbon Saving & Daily SIPP dashboards
Corrected logic	Fixed approval chain, structured Company records, template field locking, closure request/confirm/revoke flow
New capability	Assessor role & readings, two structured logs (permit + system), Global Facilities Report portal page, Announcements admin screen, gated form sections
Explicitly removed	Blocked Words feature, D1–D5 legacy dashboards, admin password reset, per-save PDF snapshots, MicronSupervisor/MicronManager/User roles, Excel export from the portal
Out of scope for this deck	Data migration planning, integration architecture, infrastructure backup/recovery — covered separately with engineering

03

User Roles & Permission Matrix

- Available roles
- Feature access per role
- Create/View/Edit/Review/Approve/Close/Cancel
- Conditional permissions
- Reviewer group management

Available User Roles

Six roles, each exclusive — nobody holds two

PROPOSED BEHAVIOUR

- Super Admin — full access to everything, including raising and approving permits (support fallback; not normal usage)
- User Admin — configuration and administration only; cannot raise, approve, or enter readings
- Micron Staff — can raise permits AND approve them (when named on a permit); logs in via SSO only
- Contractor Requestor — can raise permits for their company; cannot approve
- Contractor PM — first-stage approval for their own company's permits only; cannot raise
- Assessor — enters safety readings on assigned child permits only; no other capability at all

CONFIRM WITH END USER

? *Does any real job today not fit cleanly into one of these six roles?*

Feature Access by Role

Role	Login	Raise	Approve	Readings	Configure
Super Admin	Password	Y	Y	Y	Y
User Admin	Password	N	N	N	Y
Micron Staff	SSO	Y	Y	if assigned	N
Contractor Requestor	Password	Y	N	if assigned	N
Contractor PM	Password	N	Y (own company)	N	N
Assessor	Password	N	N	Y (assigned only)	N

MT Group Membership & Permit Assignment

Not roles — set per group or per permit

MT Group Membership

Micron Staff join ONE group (ENV, EHS, or OPS). Any one member's approval covers the whole group. Two switches per group: can close permits, can edit after final approval.

Permit Assignment

Host, Host Manager, Area Manager, and Assessor are all NAMED on a specific permit. A person in an MT Group cannot also be named an individual reviewer on the same permit.

Create / View / Edit / Review / Approve / Close / Cancel — Who Can Do What

Summary across the permit lifecycle

PROPOSED BEHAVIOUR

- Create: Super Admin, Micron Staff, Contractor Requestor (subject to the open-permit limit)
- View: scoped by involvement — Contractor sees their whole company's permits; Micron Staff sees only permits they raised or are named on; Admin sees everything
- Edit (before final approval): only the CURRENT-TURN reviewer; Admin can add/remove reviewers at any time
- Edit (after final approval): only an MT Group selected on that permit, with the edit-after-approval switch enabled — readings are never editable by anyone
- Review/Approve: whoever is currently named at the active stage of the chain
- Request Close/Cancel: only the original submitter (or Super Admin, for support)
- Confirm Close/Cancel: an MT Group selected on the permit, with the closure switch enabled

04

User Login & Account Management

- Microsoft SSO login
- Non-SSO / vendor login
- Account creation
- Activation & deactivation
- Company/role assignment

Microsoft SSO Login (Micron Staff)

First login triggers a mandatory onboarding sequence

PROPOSED BEHAVIOUR

- Azure AD confirms identity and supplies the user's email
- First-time login shows a course-check page: tick “I completed the ePTW LE course” + upload a certificate — BOTH required, or the account is not created
- Certificate is self-declared at registration; stored and viewable by Super Admin / User Admin for audit, not verified before activation
- User then supplies: Name, Email (from Azure AD), Company (fixed: Micron), Site (defaults to ALL sites, editable later by admin)
- Account created immediately as Micron Staff. Role can be changed later by an admin.

CONFIRM WITH END USER

? *Should the uploaded certificate ever require admin sign-off before the account activates, or does self-declared + auditable stay sufficient?*

Non-SSO / Vendor Login (Contractors & Assessors)

Manually created — no self-registration path exists

PROPOSED BEHAVIOUR

- Admin/Super Admin creates the account directly, via the existing manual submission process
- New account: ONE email with a link to set a password directly — no separate email-verification step, no expiry timer
- The link stays valid until the password is actually SAVED — not until the page is merely opened
- First login only: redirected once to the existing Forms & Tutorial Videos page — no new content area built for this
- From then on, logs in straight to the portal

Forgot Password & Admin Password Reset

Closing a real security gap in the current system

PROPOSED BEHAVIOUR

- Admin password reset is REMOVED entirely — today it sets a shared plaintext password with no notification and no API role check
- Forgot Password is restored to the Sign In page (currently hidden) as the only self-service password recovery route
- Forgot Password link expires after 24 hours — different from the account-creation link, since this protects an account that already exists

Activation, Deactivation & Assignment

Company and role changes cascade predictably

PROPOSED BEHAVIOUR

- Deactivating a user removes their MT Group membership automatically
- Named individual reviewer slots on LIVE permits are NOT auto-removed — flagged for manual reassignment
- Setting a Company to Inactive automatically deactivates every user under that company — one action ends a contract cleanly
- No sync with Azure AD account status — offboarding a Micron Staff leaver is fully manual

CONFIRM WITH END USER

? *Is fully manual offboarding (no Azure AD sync) acceptable, given the risk that a departed leaver stays selectable as a reviewer until someone notices?*

05

Permit Type & Template Management

- Parent permit types
- Child permit types
- Mandatory vs optional children
- Templates & checklists
- Display conditions

Parent vs Child Templates

Two separate template lists, matching how permits are actually raised

PROPOSED BEHAVIOUR

- Templates split into Parent and Child lists in the admin builder — reflects real approval-chain differences
- A template is Draft or Published; only Published templates are usable for real permits
- Field NAME auto-generated and locked forever at first publish. Field TYPE locked too.
- Label, tooltip, required/optional, and ordering remain editable at any time, even after publish
- “Delete” on a field means “hide from new permits” — never a true delete, since old permits keep referencing it

Mandatory vs Optional Child Permits

Category	Type	Behaviour
Mandatory checklist	A-series, triggered by hazard answers	Gated: ticking a hazard forces the matching child before Submit is enabled
Merging, high-risk	B1, B2, B3, B4	Raised separately; own full chain; merges into parent at COMPLETED, not at Approved
Never merging	B5 — LSS Impairment	Raised separately; own full chain; NEVER merges; has an extra Executed state

System Components — Reserved Fields

Cannot be deleted or retyped by an admin

PROPOSED BEHAVIOUR

- Start Date, End Date, Site, Discipline, Host, Host Manager, Area Manager, MT Group stages are all reserved components
- Work Description is also reserved — mandatory on every parent template, added to support the new admin keyword search (Section 16)
- These can be relabelled and marked optional, but never deleted or have their type changed
- This protects dashboards, filters, and reports from being silently broken by a template edit
- “Can be created under parent” is the ONLY merge-related setting exposed to an admin — whether a child type merges is system-determined by permit type, not editable

Conditions for Displaying / Selecting a Permit Type

Gated sections drive the approval chain automatically

PROPOSED BEHAVIOUR

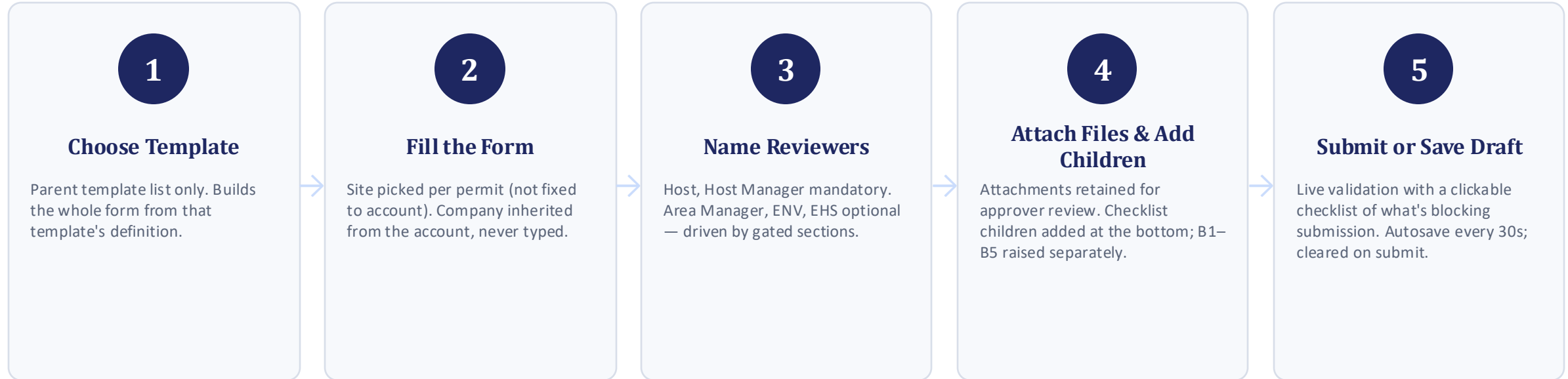
- A section header (e.g. “Environmental Impact”) must be ticked BEFORE its items can be ticked — items stay disabled until the header is active
- Ticking the header adds the matching approver (e.g. ENV Manager) to that permit's chain automatically
- Unticking the header after items were selected requires confirmation, since it clears everything inside
- B1–B5 never appear in the parent's child-permit picker — they are structurally excluded, not just hidden by a flag

06

Permit Creation Process

- Step-by-step flow
- Required/optional fields
- Requestor & company info
- Location, discipline, activity
- Attachments
- Draft & submission rules

Step-by-Step Creation Flow



Required vs Optional Fields

Mandatory reviewer stages and validation behaviour

PROPOSED BEHAVIOUR

- Mandatory: Site, Discipline, Host, Host Manager, at least one OPS-capable MT Group (closure-capable)
- Optional, form-driven: Area Manager, ENV, EHS — required only if the matching section was ticked
- All required template fields must be complete before Submit enables — checklist shown live, clickable to jump to the field
- Numeric-only fields (e.g. readings) reject non-numeric input at the field level

Draft Saving & Submission Rules

PROPOSED BEHAVIOUR

- Drafts autosave every 30 seconds; no manual save action required
- Drafts are cleared automatically once the permit is successfully submitted
- Open-permit interlock warning is shown BEFORE the form opens, not after it's filled in — states the count and which permits are blocking
- Duplicate keeps its content but clears Host and Discipline, with an explicit on-screen message explaining why

CONFIRM WITH END USER

? *Does the open-permit limit threshold need adjusting for the new system, or does it carry over unchanged?*

07

Permit Numbering Logic

- Parent format
- Child format
- Discipline/location/type codes
- Running number generation
- Number changes on merge
- Duplicate/skipped handling

Parent Permit Numbering

Format: SITECODE-TYPELETTERS-DISCIPLINE-YYYY-MM-DD/NNNN

PROPOSED BEHAVIOUR

- Example: F10A1-GB1-FAC-2026-08-18/0231
- TYPELETTERS start with the parent's own type, then gain a letter for each embedded checklist at submission, and LATER gain a letter for each B1–B4 merged in as it completes
- This means the parent's permit number CHANGES OVER TIME as children merge — G becomes GB1, then GB1B2
- Discipline and Site codes are locked once used on any permit — renaming the record afterward never changes numbers already issued

Child Permit Numbering (B1–B4)

Format: TYPELETTERS-DISCIPLINE-YYYY-MM-DD/NNNN — no site code

PROPOSED BEHAVIOUR

- Example: B1-FAC-2026-08-18/0007
- No site code — the child is linked to its parent, which already carries the site
- Once merged (at Completed), the child's own number stops being searchable — the record is deleted
- The number IS preserved permanently as text inside the parent's history log and PDF, tagged to every merged-in entry

CONFIRM WITH END USER

? *Should a merged child's number ever be made searchable again (resolving to its parent), or does “number no longer found” stay the intended behaviour?*

08

Parent and Child Permit Logic

- Parent/child relationship
- Adding/removing children
- Approval & merging logic
- Impact of child status on parent
- Independent closure/cancellation/recall

The Three Child Permit Categories

Category	Raised	Merges into parent	Closure/Cancellation
1. Checklist/embedded	Bottom of parent form, before submission	Already part of the parent	Follows the parent — no independent action
2. B1, B2, B3, B4	Separately, against an existing parent	YES — at COMPLETED only. Cancelled children never merge.	Independent — can close/cancel without affecting the parent
3. B5 — LSS	Separately, against an existing parent	NEVER — permanent standalone record	Independent — same as above, plus its own Executed state

Merge Mechanics — What Actually Happens

Fires automatically at Completed, never at Approved

PROPOSED BEHAVIOUR

- Child reaches Completed → merge fires automatically, no manual step
- Child's type letter is added to the parent's permit number
- Child's full content AND its entire log history are copied into the parent, every entry tagged with the child's original permit number
- The standalone child record is then DELETED — from this point it only exists inside the parent's PDF and log
- A CANCELLED child (any type) never merges — it remains a standalone record permanently

No Restriction Between Parent and Child Approval

A deliberate, confirmed design decision

PROPOSED BEHAVIOUR

- A separately-raised child permit (B1–B4 or B5) can reach Approved even while its parent is still Pending, PendingApproval, or Rejected
- The two chains run entirely independently — nothing links their approval progress
- This was explicitly confirmed rather than assumed — flagged here as a deliberate choice, not an oversight

CONFIRM WITH END USER

? *Confirm: should high-risk child work ever be blocked from proceeding until the parent permit itself is authorised?*

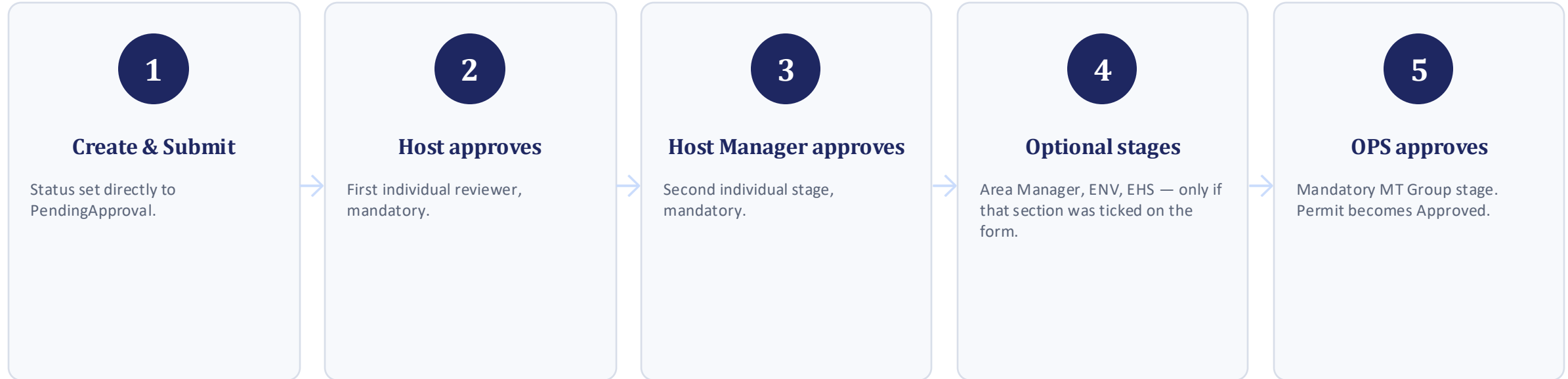
09

Permit Review & Approval Workflow

- Reviewer/approver selection
- Review stages & sequence
- Sequential (not parallel)
- Routing conditions
- Rejection & resubmission
- Delegation — considered
- Final approval

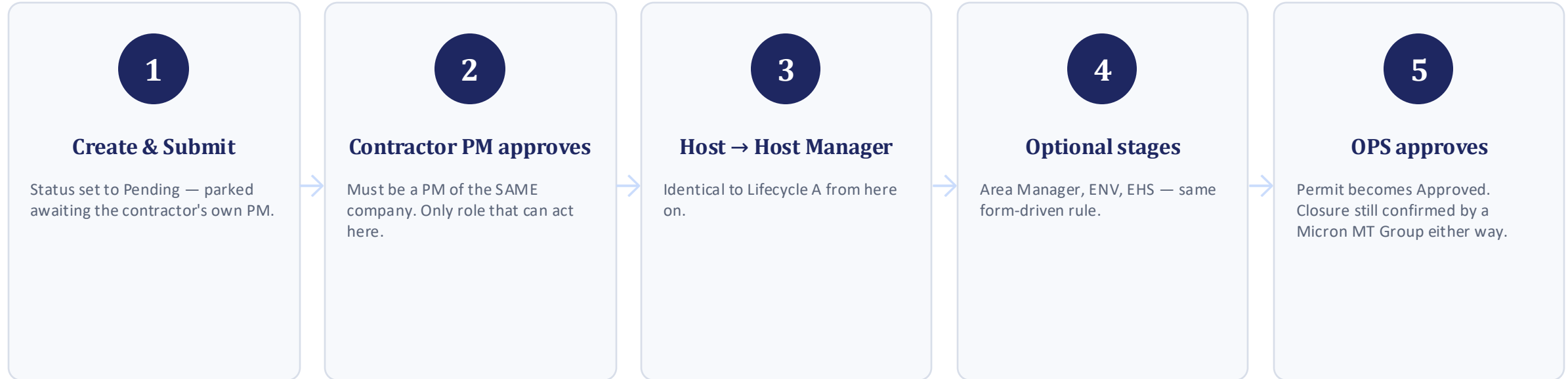
Lifecycle A — Micron Staff Permit

No PM stage — straight into the reviewer chain



Lifecycle B — Contractor Permit

One extra gate at the front; identical from there on



Sequence Is Strictly Sequential — Not Parallel

A deliberate design decision, stated explicitly for confirmation

PROPOSED BEHAVIOUR

- Individual stages act in fixed order: Host → Host Manager → Area Manager
- Then group stages, in fixed order: ENV → EHS → OPS
- One member's signature satisfies the whole group stage — but stages themselves never run at the same time
- The sequence is enforced by the SERVER, not just the screen — closing the biggest defect in the current system

CONFIRM WITH END USER

? *Parallel approval (e.g. ENV and EHS reviewed at the same time) was NOT included — confirm this is acceptable, or should any stages run in parallel to speed up the chain?*

Rejection & Resubmission

One consistent rule — no route-dependent inconsistency

PROPOSED BEHAVIOUR

- Rejection requires a mandatory reason
- Rejecting RESTARTS the whole chain from the beginning — every reviewer signs again on the corrected version
- This differs from today, where earlier approvals are sometimes kept and sometimes wiped depending on which button was used
- Distinct from editing: reject is for when the REQUESTOR must supply new information; a reviewer who can fix it themselves edits instead

CONFIRM WITH END USER

? *Restarting the full chain on every rejection means more re-signing effort — confirm this trade-off (safety vs. speed) is acceptable.*

Reviewer Reassignment — No Formal Delegation Feature

Considered and deliberately excluded

PROPOSED BEHAVIOUR

- MT Group stages need no delegation — any member of the group can act, which already covers someone being on leave
- For named individuals (Host, Host Manager, Area Manager): a reviewer or admin can remove them and add a replacement at any time, fully logged
- A formal “nominate a backup in advance” feature was considered and excluded — manual reassignment was judged sufficient

CONFIRM WITH END USER

? *Confirm manual reassignment is sufficient for named-individual absences, or is a formal delegation feature still needed?*

Final Approval Conditions

PROPOSED BEHAVIOUR

- A permit becomes Approved the moment the LAST stage in its chain signs — whether that's OPS, or an earlier stage if later ones weren't triggered
- Every permit must carry at least one closure-capable MT Group — checked and required at creation, so a permit can never become unclosable
- Approval timestamp (approveAt) is a permanent fact once set — used later for reporting regardless of what happens to the permit afterward

10

Permit Status & Lifecycle

- Draft
- Pending review/approval
- Approved
- Rejected/returned
- Pending closure
- Completed
- Pending cancellation
- Cancelled
- Status transitions

All Statuses

No separate 'Expired' status — see Section 13

Status	Meaning	Reached by
Draft	Saved but not submitted	Save as Draft
Pending	Awaiting Contractor PM (contractor only)	Submitted by a non-Micron company
PendingApproval	In the reviewer chain	Submitted by Micron staff, or after the PM approves
Rejected	Sent back with a reason	Any approver rejects
Approved	Fully signed off	The last chain stage signs
Executed (LSS only)	Life-safety system currently impaired	MT Group presses Execute on an approved LSS permit
PendingClosure	Closure acknowledgement submitted, awaiting confirmation	Submitter/Host/Contractor signs the closure section
PendingCancel	Cancellation acknowledgement submitted, awaiting confirmation	Same acknowledgement section, cancellation path
Completed	Closed out — end state	Confirmed by a closure-capable MT Group, or LSS Complete pressed
Cancelled	Cancelled — end state	Confirmed by a closure-capable MT Group

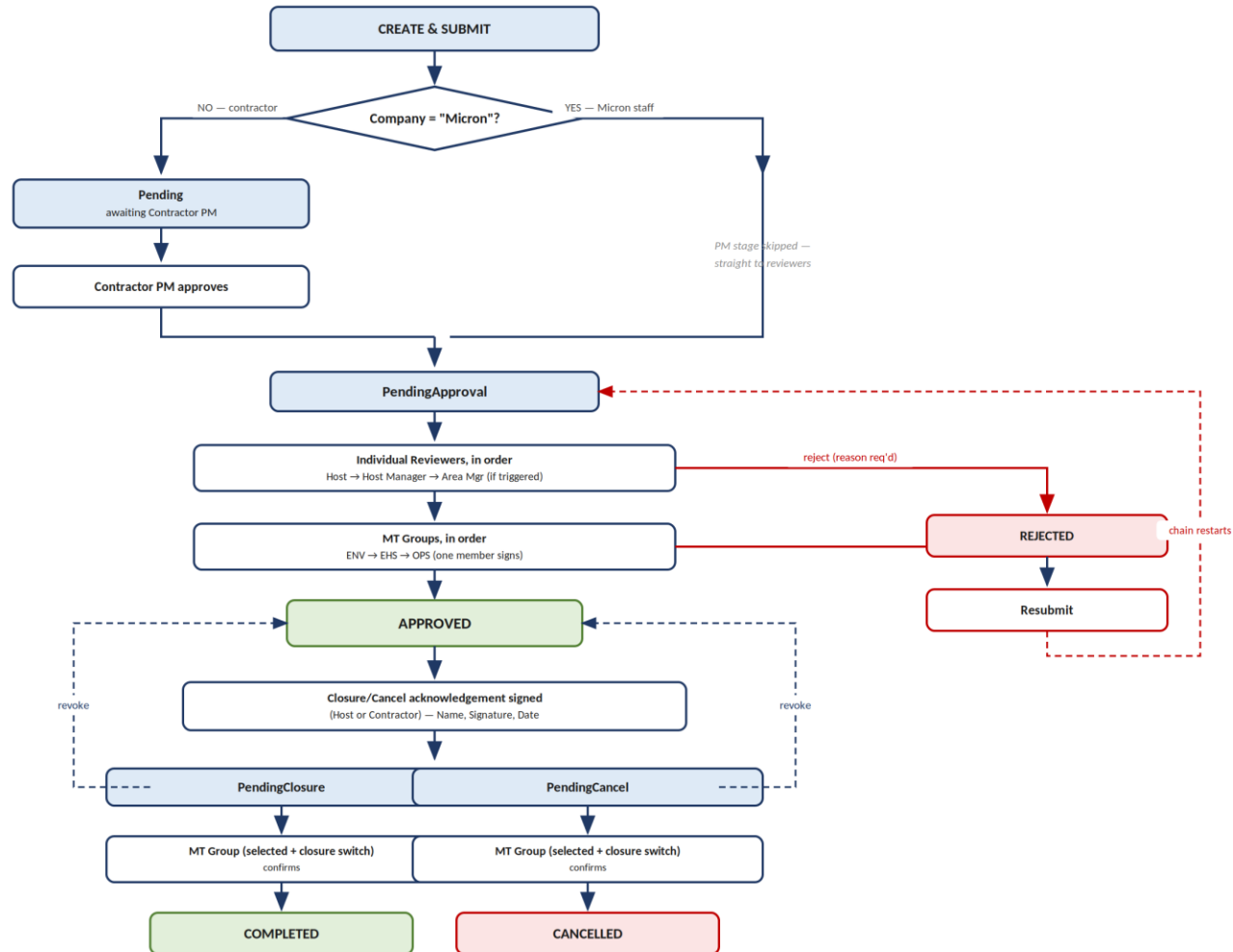
Status Transition Rules

PROPOSED BEHAVIOUR

- Rejection always returns to the very start of the chain — never resumes mid-chain
- Recall undoes exactly ONE step — the most recent approval only, never further back
- Revoke returns PendingClosure/PendingCancel back to Approved
- No automatic “Expired” transition exists — an overdue Approved permit simply stays Approved (see Section 13)

Full Lifecycle — Parent Permit

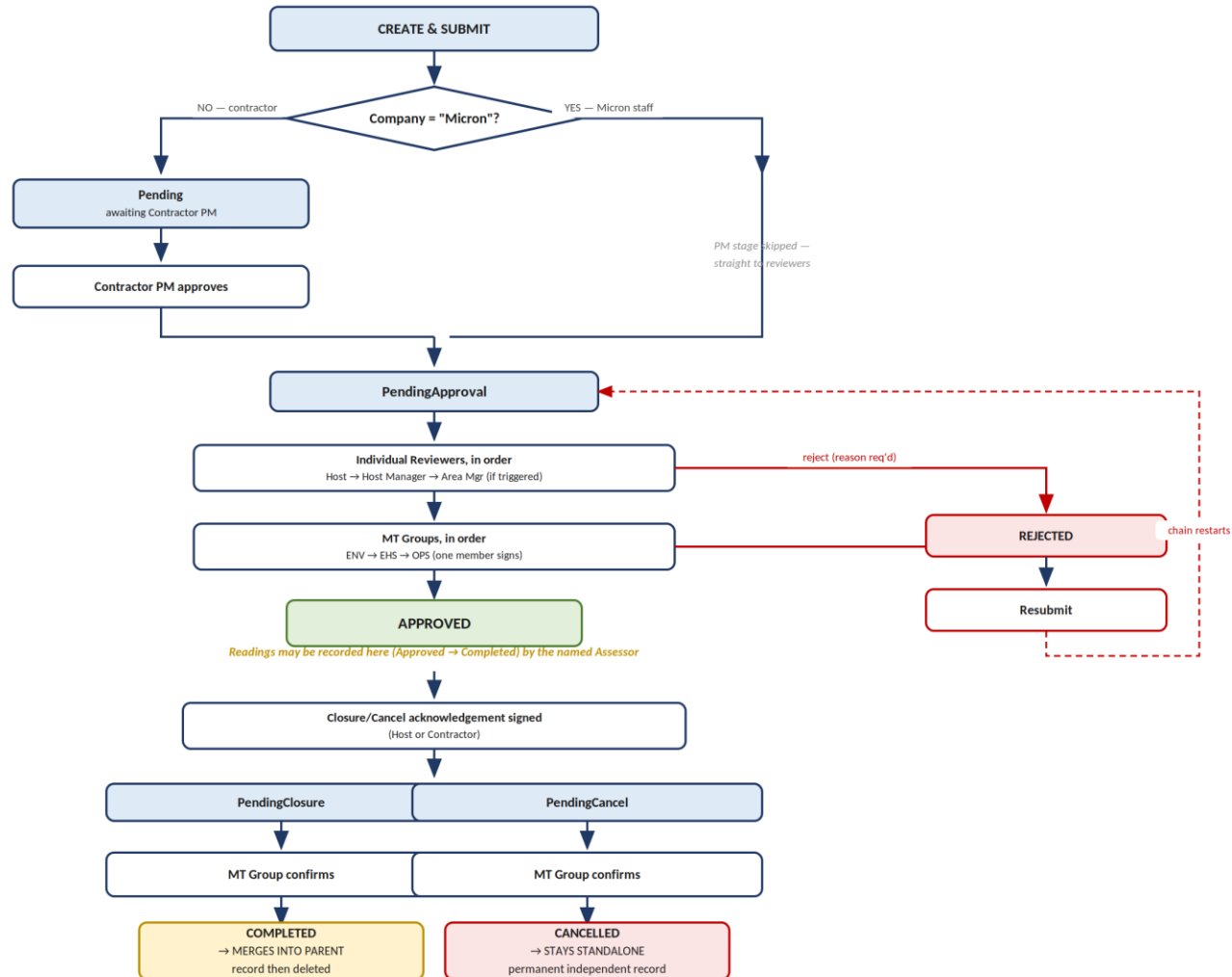
Creation through closure, including the reviewer chain and reject loop



Note: B1-B4 child permits merge into this parent as each completes — the permit number gains a letter each time.

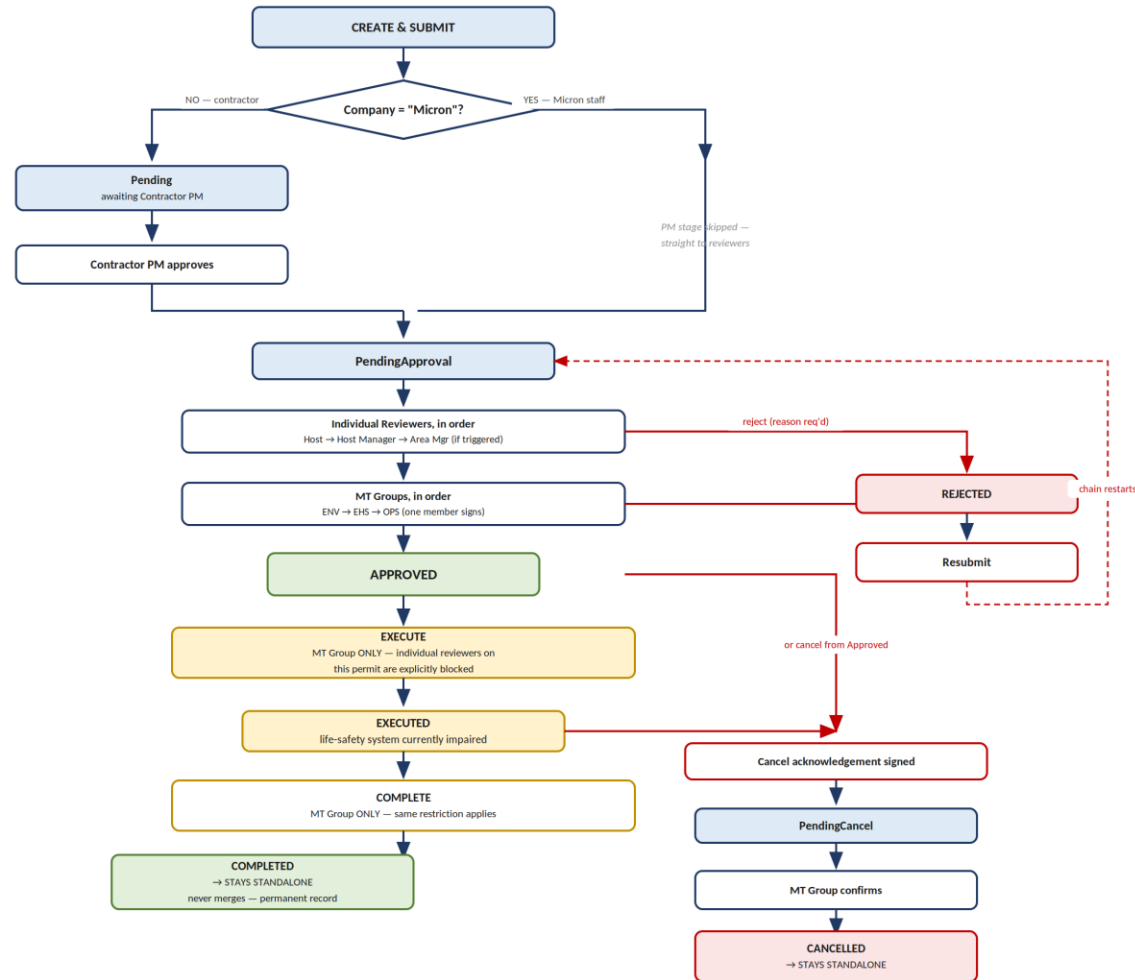
Full Lifecycle — Child Permit (B1-B4)

Same chain as the parent, but Completed MERGES in while Cancelled stays standalone



Full Lifecycle — LSS Permit (B5)

Adds Execute/Complete, restricted to MT Group only — never merges into the parent



Note: LSS never merges into its parent, regardless of outcome — a permanent, non-configurable rule.

11

Field & Button Enable/Disable Logic

- Buttons per status
- Requestor-editable fields
- Reviewer-editable fields
- Locking after approval
- Special edit permission

Who Can Edit, and When

Stage	Who can edit content	What they can touch
Before submission	The requestor	Everything
During review (not yet final approval)	Only the CURRENT-TURN reviewer	Full content, including dates
After final approval	Only an MT Group selected on the permit, with the edit-after-approval switch enabled	Full content EXCEPT readings, which nobody can ever edit
Admin, at any time	Super Admin / User Admin	Reviewer list only (add/remove) — never permit content directly

Button Availability by Status

Actions appear only when they're actually valid

PROPOSED BEHAVIOUR

- Approve / Reject: visible only to the current-turn reviewer; hidden once the permit is already approved, rejected, or the user is the submitter (self-approval blocked)
- Request Closure/Cancellation: visible only to the original submitter, only when status is Approved
- Confirm Closure/Cancellation: visible only to an MT Group member selected on that permit with the closure switch enabled, only when status is PendingClosure/PendingCancel
- Revoke: visible to the submitter or Super Admin, only while a closure/cancellation request is still pending confirmation
- Recall: visible only to the specific approver who gave the most recent approval, or Super Admin
- Duplicate: visible only to the original requestor, hidden for Super Admin entirely

New: Closure/Cancellation Acknowledgement Interlock

A signed declaration that IS the closure/cancellation request

PROPOSED BEHAVIOUR

- A dedicated acknowledgement section (Name, drawn Signature, Date) sits greyed out and unfillable for the ENTIRE life of the permit up to Approved
- Once Approved, the section unlocks — either the Contractor or the Micron Host (whichever applies) can complete it
- Submitting this section IS the act of requesting closure/cancellation — status moves to PendingClosure/PendingCancel at that moment
- Name, Signature, and Date lock PERMANENTLY the instant it's submitted — frozen even from the edit-after-approval MT Group
- This section is shared by both closure and cancellation — same fields, same lock rule
- The MT Group Confirm step still happens afterward, unchanged, as a separate action

CONFIRM WITH END USER

? *If a closure/cancellation request is revoked, should this signed acknowledgement be cleared for a fresh signature, or does the original signed record stay on file even through a revoke?*

12

Approved Permit Amendment

- Content locked after approval
- Who can amend
- Group membership requirement
- Fields allowed to change
- Reapproval requirement
- Audit history

Amending an Approved Permit

One MT Group switch controls all post-approval editing

PROPOSED BEHAVIOUR

- Content is locked to everyone EXCEPT an MT Group that is (a) selected as an approver on that specific permit, AND (b) has the edit-after-approval switch enabled
- Both conditions must hold — group membership alone, without being selected on that permit, is not enough
- Fields allowed to change: any permit content, including start/end dates — EXCEPT the readings table, which is never editable by anyone once saved
- No reapproval is triggered by a post-approval edit — the change takes effect silently on the existing approved permit
- Every edit is logged with old value, new value, who, and when — but nobody is notified automatically

CONFIRM WITH END USER

? *Should certain high-impact edits (e.g. extending the end date by more than X days) trigger a notification to the original approvers, even though the edit itself doesn't require reapproval?*

13

Permit Extension & Expiry

- Validity period
- Start/end date control
- Extension process
- Expiry notification
- Automatic expiry logic

No New Expiry Mechanism — Extension and Expiry Are Handled by Existing Rules

This section resolves to “no change needed” after review

PROPOSED BEHAVIOUR

- There is NO separate ‘Expired’ status. A permit past its end date keeps its current status (e.g. stays Approved)
- “Extension” is simply an edit to the end date — covered entirely by the edit-after-approval MT Group switch in Section 12
- An overdue, unclosed permit continues to appear normally in listings and dashboards — no label, icon, colour, or notification marks it as overdue
- The ONLY consequence: an overdue Approved permit still counts toward the existing open-permit interlock, exactly as any other open permit does — there is no separate overdue threshold or configurable grace period

CONFIRM WITH END USER

? *Confirm this is sufficient — no dedicated Expired status, no overdue flag, no expiry reminder email. Extension and overdue handling both ride on rules already built for other purposes.*

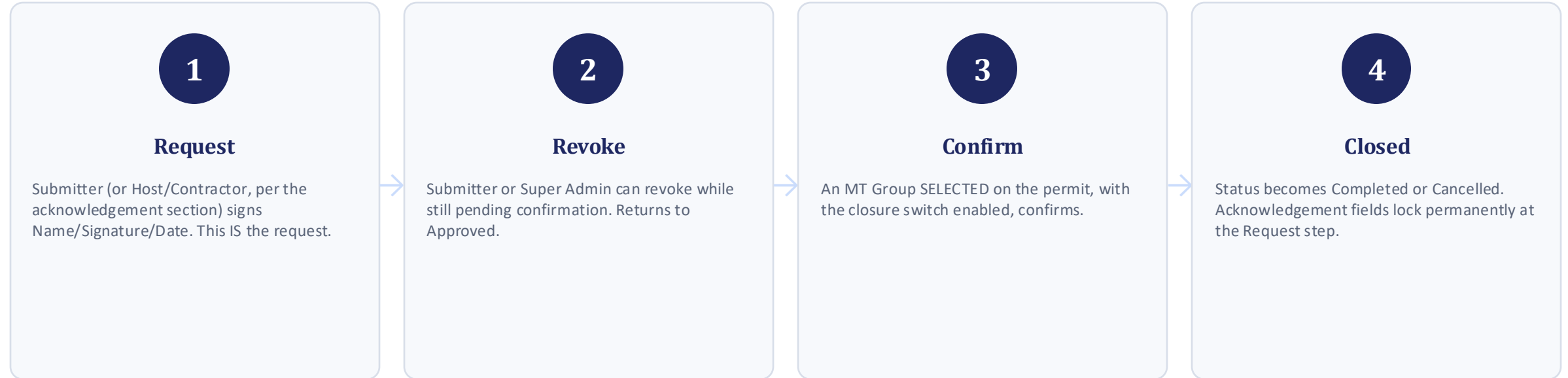
14

Permit Closure, Cancellation & Revocation

- Who can request
- Who approves
- Withdrawing/revoking
- Remarks & attachments
- Impact on child permits

Closing a Permit

Request → Revoke → Confirm → Closed



Impact on Related Child Permits

Parent and child permits close independently

PROPOSED BEHAVIOUR

- A separately-raised child permit (B1–B4, or LSS) can be closed or cancelled independently of its parent's status
- A child reaching Completed still merges into the parent, exactly as normal — closure and merge are the same event for B1–B4
- A child reaching Cancelled does NOT merge — stays a standalone record permanently
- LSS never merges regardless of outcome (Completed or Cancelled) — this is a permanent, non-configurable rule

CONFIRM WITH END USER

? *Who should be authorised to change an LSS permit to Executed, and to Closed / Cancelled — see Section 24, item 16*

15

Email & System Notifications

- Submission
- Review/approval
- Rejection
- Closure/cancellation
- No escalation — by design
- Recipients

All Notifications

Email only — no in-app inbox, no push, no deep links

Event	Sent to
Permit submitted	Requestor
Awaiting PM approval (contractor)	Contractor PMs of that company
Your turn to review	Next reviewer / MT Group
Approved (final)	Requestor
Rejected	Requestor, with reason
Closure/cancellation REQUESTED	The closure-capable MT Group on that permit (new)
Closure/completion confirmed	Requestor
Cancellation confirmed	Requestor
New account created	New user
Assigned as assessor	The assessor (new)

No Escalation or Reminders — A Deliberate Exclusion

Considered and explicitly decided against

PROPOSED BEHAVIOUR

- No reminder is sent if a permit sits unactioned at a reviewer's stage for any length of time
- No escalation to a manager or admin after a threshold period
- If something stalls, following up is the requestor's responsibility, same as today
- This was a deliberate simplicity choice — flagged here since it's a common expectation for a permit system

CONFIRM WITH END USER

? *Confirm no escalation/reminder mechanism is acceptable — or is this needed, at least for the closure request stage, which can otherwise sit indefinitely?*

16

Permit Search & List Management

- Searchable info
- Filtering
- Sorting/pagination
- Role-based visibility
- Export

Visibility Rules

Scoped strictly by involvement — a real tightening from today

PROPOSED BEHAVIOUR

- Contractor Requestor / PM: sees every permit belonging to their OWN COMPANY (not just their own submissions)
- Micron Staff: sees ONLY permits they personally raised, or where they are named as a reviewer/approver/assessor — no browse-everything view
- Super Admin / User Admin: sees everything
- One listing page with tabs that adapt automatically per role — no empty tabs shown for capabilities a role doesn't have

CONFIRM WITH END USER

? *Micron Staff losing a general 'browse all permits' view is a real behaviour change from today — confirm this tightened visibility is acceptable operationally.*

Listing Columns — Parent Permit

13 columns, search or filter as marked

#	Column	Interaction
1	Permit Number	Search
2	Permit Code	Search
3	Permit Type	Filter
4	Discipline	Filter
5	Requestor Email	Search
6	Requestor's Company	Filter
7	Host	Search
8	Applier Site	Filter
9	Date Applied	Filter
10	Start Date	Filter
11	End Date	Filter
12	Status	Filter
13	Action	View / Review (if your turn) / Export PDF / Duplicate (own permits)

Listing Columns — Child Permit (B1–B4) & LSS

Child listing links back to its parent; LSS gets its own dedicated day-by-day view

Child Permit (B1–B4) — 12 columns

Child Type
Child Permit Number
Parent Permit Number
Discipline
Requestor Email
Requestor Company
Host
Date Applied
Start Date
End Date
Status
Action — View / Review / Export PDF / Duplicate

LSS (B5) — dedicated view

Day-by-day grouping, sub-grouped by status:
Pending · Approved · Executed · Completed
15-day default window (7 before, today, 7 after)
Filters: Site, Date range
Click a count → popup listing the
contributing permit numbers

Export & Sorting

PROPOSED BEHAVIOUR

- Sort: newest first, by creation date
- Page size: user-selectable
- No bulk actions in the listing — it is purely a find-and-open tool
- No Excel export from the portal listing — the only Excel output is the separate Global Facilities Report (Section 17)

New: Search Permits — Admin Keyword Search

A separate, dedicated page — distinct from the tabbed listing above

PROPOSED BEHAVIOUR

- New nav item, separate from the main permit listing — Super Admin and User Admin only
- Scope: parent permits only — searches across EVERY parent permit, with no company or involvement scoping
- All matching is PARTIAL (“contains”), consistently across every field
- Filters: Permit Code, Work Description, Creation Date range, Host Email, Requestor Company, Discipline
- Work Description becomes a new RESERVED system component — mandatory on every parent template, name locked, cannot be removed by an admin — this is what makes it reliably searchable

Search Results — Columns

10 columns; Permit Number is clickable and opens the permit page

#	Column
1	Permit Number (click to open)
2	Requestor Name
3	Requestor Company
4	Host Email
5	Discipline
6	Creation Date
7	Start Time
8	End Time
9	Work Description
10	Status

17

Dashboards & Reports

- Operational dashboard
- Monthly permit report
- Approved/completed counts
- Report calculation logic
- Excel/PDF export
- Access permissions

Dashboards — Six Kept or Rebuilt, Everything Else Removed

D1–D5 and the legacy V1 dashboard are DROPPED entirely — internally inconsistent, never reconciled with each other

PROPOSED BEHAVIOUR

- Kept as-is: Carbon/Paper Saving, Daily SIPP Works
- New: Monthly Approved Permit Count, Child Permit Count by Type, General vs Tool Install, Child Permit Tree (drill-down)
- Every dashboard is filtered and calculated consistently — no two reports silently use different date fields or status sets

The Six Dashboards — Calculation Logic

Dashboard	Date basis	Status scope
Carbon / Paper Saving	Work start date	All statuses
Daily SIPP Works	Child start date	Approved, PendingApproval
Monthly Approved Permit Count	approveAt (locked permanently once set)	Any — count survives later status changes
Child Permit Count by Type	Creation date	Child's own status: Approved, PendingClosure, Completed
General vs Tool Install	Creation date	All statuses
Child Permit Tree (drill-down)	Creation date	All 7 statuses

Global Facilities Report — Scheduled Excel Export

A backend job, not a portal screen — kept from the current system

PROPOSED BEHAVIOUR

- Generated automatically every day at 9:00am, unchanged from today
- Uploaded to S3 using the SAME FILENAME every day — this is a live pointer to today's data, not a dated archive
- Content structurally similar to today: every permit flattened by risk category, with per-role signature status
- NEW: a dedicated portal page with a download link and “Last generated: [date, time]” — replaces relying on a bookmarked raw S3 URL
- Access restricted to Super Admin and User Admin only

CONFIRM WITH END USER

? *Does the report's column set / risk-category logic need updating for the new data model, or does “structurally similar to today” fully cover it?*

18

Permit PDF & Document Generation

- PDF content
- Parent/child presentation
- Checklists & signatures
- Approval history
- Regeneration after amendment

What the PDF Contains

One document, generated live, always complete

PROPOSED BEHAVIOUR

- Generated ON DEMAND from current state — no stored per-save snapshots, unlike today's mechanism
- Anyone who can view the permit can generate it, at any time
- ALL merged children's full content appears inside the parent PDF — there is no separate child PDF
- Attachments included as clickable links, not embedded files
- Complete history log at the end — every entry, chronological, tagged by source (parent or which merged child)
- No table of contents or bookmarks — it's a point-in-time snapshot, not a navigable reference document

Regeneration & Identification

PROPOSED BEHAVIOUR

- Filename = parent permit code + generation timestamp — since the same permit can look different at different points in time as children merge
- No manual “regenerate” action needed — every generation reflects current state automatically
- A generated-timestamp line appears inside the document itself, so two printed copies of the same permit can be told apart
- No QR code or other verification marker is included on the PDF — the generation timestamp is sufficient

19

Audit Trail & Change History

- Creation/update history
- Review/approval records
- Field-level tracking
- User/date/time
- Amendment reasons

Two Logs, Two Purposes

	Permit Log	System (Configuration) Log
Scope	One log per permit	System-wide admin activity
Where	Inside the permit's own page	Dedicated admin page (not per-object)
Visible to	Anyone who can see the permit	Super Admin and User Admin only
Captures	Created, edited (old/new value), approved, rejected, recalled, readings, merges (tagged with child's number), failed attempts	Templates, users, sites, MT Groups, companies, disciplines, maps
Entry grouping	One entry per action	One entry per SAVE — all changed fields grouped, not one entry per field

Structure, Timezone & Retention

PROPOSED BEHAVIOUR

- Every entry: actor ID + name at the time, event type, field, old value, new value, outcome (success/failed with error message)
- Timestamp stored in UTC internally, always DISPLAYED in Singapore time
- Both logs are append-only — never edited or deleted by anyone, including admins
- Retention: kept forever, no purging, ever
- Filterable by event type, source, and actor
- Login/logout events are NOT captured in the system log — considered and excluded

20

Administration & Configuration

- User & role admin
- Reviewer group setup
- Company & contractor mgmt
- Site & discipline config
- Template config
- Announcements

Sites, Maps & Disciplines

Full admin control, with a real security gap closed

PROPOSED BEHAVIOUR

- All three: full Create/Edit by admin — Sites currently have NO edit screen at all today
- Soft delete ONLY — an Active/Inactive toggle, never a true delete; anything referencing an inactive record keeps working exactly as before
- Codes enforced unique, and LOCKED once used on any permit
- Permits display the NAME everywhere — code is only ever used inside the formatted permit number itself
- Discipline create/update currently has NO permission check in the API at all — closed in the rebuild

Company Management

Replaces a free-text field that silently strands permits today

PROPOSED BEHAVIOUR

- Company becomes a real record: Name, Code, Type (Internal/Contractor), Status
- Micron itself is a Company record too, flagged Internal — the whole Micron-vs-contractor lifecycle branch becomes a flag check, not a string comparison
- Company code enforced unique
- Setting a Company Inactive cascades to deactivate every user under it

MT Group / Reviewer Group Setup

Membership, not a role — with two configurable switches

PROPOSED BEHAVIOUR

- One group per person — a person is either a group approver, or an individual reviewer (Host/Host Manager/Area Manager), never both
- Two switches per group: Can Close Permits, Can Edit After Approval
- Adding/removing members from a group updates every LIVE permit using that group automatically

Announcements — New Admin Feature

From a database-only edit to a proper admin screen

PROPOSED BEHAVIOUR

- Rich-text pop-up shown on the Sign In page, before login
- Multiple active announcements shown as bullet points inside ONE pop-up
- Always-on until manually turned off — no scheduling, shown every login with no dismissal memory
- Managed by Super Admin and User Admin

21

Security & System Controls

- Role-based access control
- Company/site data separation
- Session & link expiry
- Document access control
- Backup/recovery — infra scope

Security Controls in Scope for This Review

PROPOSED BEHAVIOUR

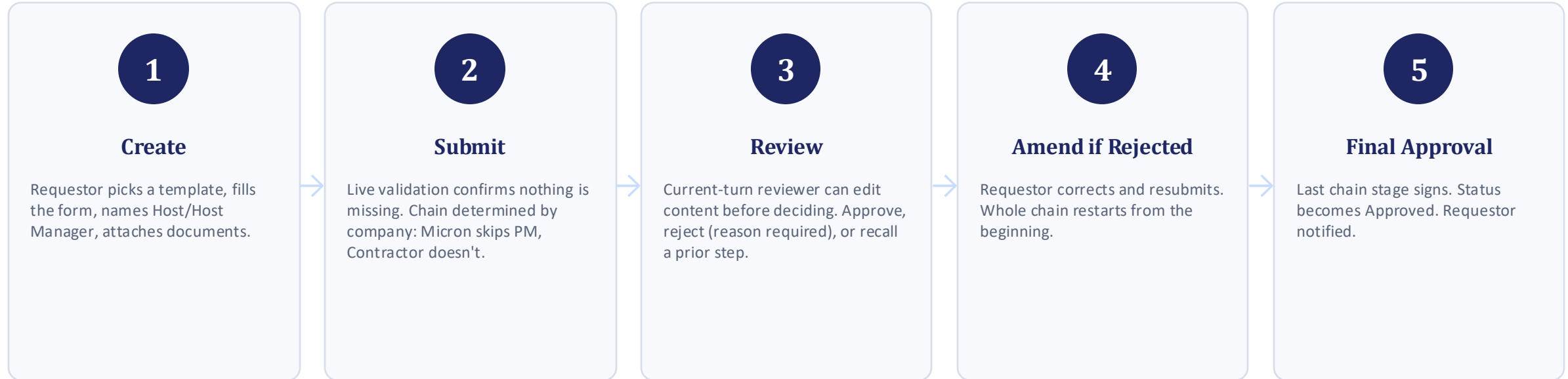
- Role-based access is enforced server-side for every action, not just hidden in the UI — closes the current system's biggest gap (approval sequence enforced only by the screen)
- Contractor visibility strictly scoped to their own company's data — no cross-company visibility possible
- Account-creation links: no expiry, valid until the password is set. Forgot-password links: 24-hour expiry
- Attachments and PDFs are only accessible to users who can already see the permit they belong to
- Backup, recovery, and infrastructure-level controls are designed separately as an engineering/IT concern — out of scope for this workshop

22

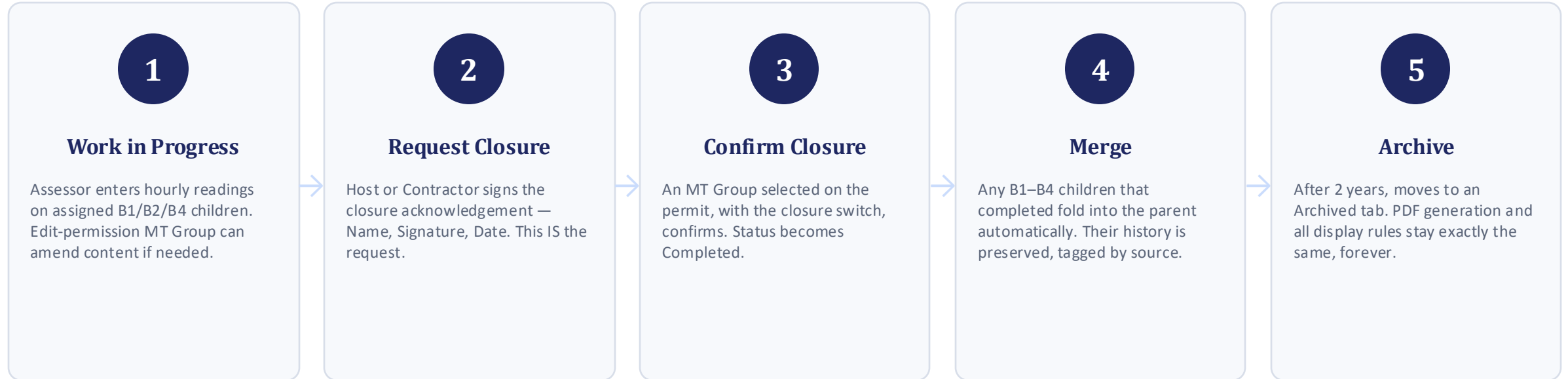
Proposed User Journey Demonstration

- Create a permit
- Submit for review
- Review & amend
- Final approval
- Manage an approved permit
- Request closure
- Complete & archive

End-to-End Journey — Part 1: Creation to Approval



End-to-End Journey — Part 2: Work to Archive



23

Key Changes from Existing ePTW

- Retained behaviour
- Corrected logic
- New features
- Processes removed
- Impact on current users

What's Being Removed Entirely

Named plainly, so nobody is surprised when it's gone

Removed	Why
Blocked Words feature	Obsolete — Micron Manager becomes a mandatory selected reviewer, not free text to police
Dashboards D1–D5 + legacy V1	Internally inconsistent, replaced by six purposeful reports
Admin password reset	Insecure workaround, replaced by a working Forgot Password
Per-save PDF snapshot mechanism	Replaced by on-demand generation from current state
MicronSupervisor, MicronManager, MTGroup-as-role, User roles	Collapsed into six roles with genuinely distinct capabilities
Free-text company field	Replaced by a managed Company record
Portal Excel export	Unreachable and unused — only the separate Global Facilities Report remains

Impact on Current Users

What people will notice most

PROPOSED BEHAVIOUR

- Micron Staff: lose the ability to browse every permit — visibility now strictly follows involvement
- Requestors: raising a permit is stricter (mandatory Host/Host Manager, gated sections) but gives clearer submission feedback
- Approvers: must now sign again on ANY rejection-resubmission, even a small fix — no partial chain resumption
- Everyone: onboarding is different — Micron Staff face a course-completion gate; contractors see a one-time tutorial
- Admins: two new admin screens (Announcements, Global Facilities Report page); one screen removed (admin password reset)

24

Items Requiring End-User Confirmation

Every open question from this review, collected in one place

Open Questions — Part 1

#	Question	Section
1	Does any real job today not fit cleanly into one of the six roles?	3
2	Should the LE course certificate require admin sign-off before activation, or stay self-declared?	4
3	Is fully manual offboarding (no Azure AD sync) acceptable?	4
4	Does the open-permit limit threshold need adjusting for the new system?	6
5	Should a merged child's permit number ever become searchable again?	7
6	Should high-risk child work ever be blocked until the parent itself is authorised?	8
7	Should any stages run in parallel rather than strictly sequential?	9
8	Is restarting the full chain on every rejection an acceptable trade-off?	9
9	Is manual reviewer reassignment sufficient, or is formal delegation still needed?	9

Open Questions — Part 2

#	Question	Section
10	Should a revoked closure/cancellation clear the signed acknowledgement, or keep the original record?	11
11	Should high-impact edits (e.g. large date extensions) trigger a notification even without reapproval?	12
12	Confirm no dedicated Expired status and no overdue flag/reminder is needed.	13
13	Should escalation/reminders exist at least for closure requests specifically?	15
14	Confirm tightened Micron Staff visibility (no browse-all view) is operationally acceptable.	16
15	Does the Global Facilities Report's column set need updating for the new data model?	17
16	Who should be the person to change the LSS permit to Executed, and to Closed / Cancelled?	8